

Usman Akinyemi

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EDUCATION

BTech Computer Science and Artificial Intelligence	Plaksha University, India	2026
Software Engineering Program	Holberton, ALX Africa	Jan 2023 - 2024

EXPERIENCE

Software Engineer (Outreachy) [🔗](#) C, Bash, Git **Git (OpenSource)** Dec 2024 - Mar 2025

- Enhanced **Git v2 protocol** by extending the **agent string** capability to include **operating system details**, improving **debugging**, **security monitoring**, and **upgrade recommendations**.
- Implemented **OS detection** using the **uname(2) system call**, ensuring **seamless integration** without breaking **backward compatibility** and handling **platform-specific variations**, particularly for **Windows**.
- Refactored **Git's codebase** to reduce reliance on **global variables** by removing the **_repository** and replacing it with **explicit repository arguments**, improving **modularity**, **maintainability**, and **testability**.

Data Engineering Intern - Python, Kafka, Docker **Dixon Lab, Plaksha University** May 2024 - July 2024

- Configured a secure **multi-broker Kafka cluster** with **SSL/TLS** and **SASL/SCRAM**, enhancing data security and compliance.
- Configured **Kafka topics** to stream **FTP server files** and **MQTT data** (temperature, pressure, CPU/RAM) to **Cassandra**, **PostgreSQL**, and **local systems** using **Strimzi connectors** and Python consumer and producer.
- Implemented **Kafka topics** for **seamless video streaming** from **IoT nodes** to **local systems**.
- Researched and presented **optimal Kafka deployment strategies** to enhance system performance.

SDE Intern [🔗](#) C, Linux, Bash, Mkosi, Git, Linux **Systemd (OpenSource)** Mar 2024 - April 2024

- Developed a **new unit test framework** with assertion macros which enhanced debugging by providing **detailed error reports** with file names, line numbers, and expression values upon failure, **improving issue identification** and **resolution**.
- Updated approximately **22 existing unit test files** by modifying **403 lines of code** to incorporate the new assertion macros, resulting in **improved logging details** and enhanced **overall test coverage** and **debugging efficiency**.
- Implemented the **--json** option for the **bootctl status** command and updated the **integration tests**, enabling machine-readable JSON output for **comprehensive bootloader status information**.

PROJECTS

SuperVaani - TypeScript, LangGraph, MySQL, Flask **ILGC, Plaksha University** Jan 2024 - Present

- Developed a **RAG application** using **LangChain LangGraph** and a **text-to-SQL** model to efficiently handle queries from the Plaksha University community, leveraging **open-source LLM (Llama 3.1)** for **data privacy** and Langsmith for model testing.
- Built a **chatbot interface** with React and TypeScript, and created **API endpoints** using flask for RAG model Integration.
- Deployed the RAG application on **Linux server** using **HAProxy** for SSL, **Nginx** as a web server, and **Gunicorn** as a app server.

OpenSource Contributor [🔗](#) Python, Docker **Open Science Initiative for Perfusion Imaging** Mar 2024 - Present

- Added a **command-line interface** to the existing 4D IVIM phantoms generator, with **detailed documentation** for usage.
- Created a Python script for efficient reading and writing of NIfTI images, improving **data processing workflows**.
- Dockerized** the **TF2.4_IVIM-MRI_CodeCollection** project and implemented a **GitHub Action** for **automated Docker image building** and **testing**, ensuring **consistent deployment** and a streamlined **CI/CD pipeline**.

Dengue Prediction Model [🔗](#) **Machine Learning Project, Plaksha University** May 2024 - July 2024

- Developed a **dengue fever predictive model** with an **MAE of 25.31**, outperforming **78%** of models on DrivenData.
- Applied **PCA** and **data preprocessing** (interpolation, normalization, log transformation), reducing model error by 5%.

FrontEnd Developer [🔗](#) HTML, CSS, JavaScript, Fetch API **DesignIT Africa** July 2024

- Built a complete, interactive, and responsive website, translating Figma designs into a functional site with **custom blog** features.
- Integrated **API endpoints** using **Fetch API** for real-time data retrieval and updates, enhancing website functionality.

SKILLS

General-Purpose Programming:	C, C++, Python, Bash Scripting, SQL
Web-Development:	HTML, CSS, JavaScript, ReactJS, TypeScript, NextJS, Flask, RestAPI, MySQL.
ML/AI:	Numpy, Pandas, Matplotlib, Scipy, PyTorch.
Tools/DevOps:	Linux, Kafka, Git/Github, Github Actions, Nginx, Haproxy, Docker, Puppet.

Other OpenSource Contribution

- Official Python Documentation** - Contributed to improving Python Official Documentation.
- LLVM Project** - Fixed a bug in Clang's Extract API for Objective-C JSON generation and optimized the test suite in LLVM.
- Canonical Website** - Enhanced Sphinx documentation by integrating GitHub contributor information via the GitHub API.
- pep8speaks** - Ensured Python 3.9+ compatibility by updating Mapping imports.
- LibreOffice** - Fixed bugs by removing unnecessary **const_cast**, improving **const-correctness**, **readability**, and **safety**.